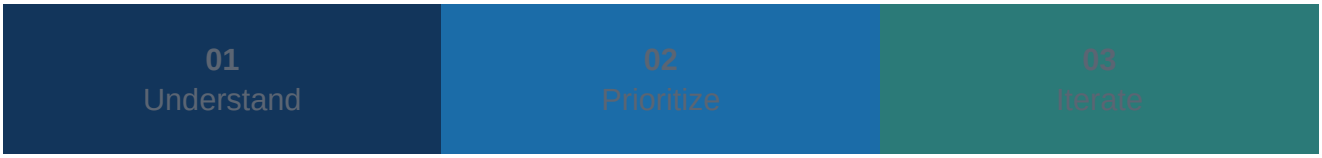
UNIT 2 • AGILE MINI PROJECT

Enterprise Student Record and Academic Management Platform

Agile Product Planning • Scrum • Epics • User Stories • Backlog • Sprints



Prepared from the supplied Unit-2 Agile Mini-Project assignment. Terminology and required activities follow the source document.

1. Project Context & Unit-2 Objective

Unit 2 translates the Unit-1 Product Discovery work into Agile planning deliverables for the Student Record and Academic Management Platform. The team reviews the Product Vision, Mission, stakeholders, Business Requirements, Functional and Non-functional Requirements, User Personas, MVP and Product Roadmap before translating them into Agile deliverables.

The goal is to establish a prioritized, incremental development plan before detailed implementation begins.

1.1 Unit-1 Inputs

• Product Vision and Mission Statement

• Stakeholder analysis and user personas

• Business, functional and non-functional requirements

• Minimum Viable Product (MVP)

• Product roadmap and project scope

2. Agile Product Vision

Product Vision: Develop a secure and user-friendly Student Record and Academic Management Platform that enables educational institutions to efficiently manage student records, attendance, academic performance, course registration, and reporting through a scalable web-based application.

2.1 Product Objective

• Centralize academic information into a structured platform.

• Enable students and faculty to access or update relevant academic information.

• Support administrators in managing academic records and reports.

• Deliver functionality incrementally through prioritized Agile work.

2.2 Primary Users

• Students

• Faculty Members

• Academic Administrators

• Examination Cell Staff

• IT Support Personnel

2.3 Business Benefits

• Improved organization and accessibility of student records.

• More systematic attendance and academic-performance management.

• Better support for course and result workflows.

• Faster creation of academic reports.

• A scalable foundation for future development.

3. Scrum Team Structure

The assignment requires teams to organize themselves into Scrum Teams and document responsibilities for the Product Owner, Scrum Master and Development Team.

Role	Primary responsibility	Project-specific responsibilities
Product Owner	Maximize product value	Define requirements; prioritize Product Backlog; represent stakeholder interests; accept completed work.

Role	Primary responsibility	Project-specific responsibilities
Scrum Master	Enable effective Scrum	Facilitate Scrum ceremonies; remove impediments; support Agile practices and collaboration.
Development Team	Build the increment	Design the application; write code; test software; prepare documentation; demonstrate completed features.

4. Epic Definition

The application is organized into high-level functional modules called Epics. Each Epic groups related capabilities so the Product Backlog can be planned and delivered incrementally.

Epic	Purpose	Key features
E1 — Student Management	Manage student lifecycle information.	Registration; Profile Management; Search; Update; Delete
E2 — Course Management	Manage course information and assignments.	Add Course; Update Course; Assign Students; View Details
E3 — Attendance Management	Record and monitor attendance.	Record Attendance; View Attendance; Attendance Reports
E4 — Marks & Academic Performance	Manage academic performance.	Enter Marks; Update Marks; Calculate Grades; Result Sheets
E5 — Authentication & Security	Protect access to the platform.	Login; Logout; Password Reset; Role-Based Access Control
E6 — Notifications	Provide timely academic alerts.	Attendance Alerts; Examination Notifications; Result Notifications
E7 — Reporting	Generate academic and operational reports.	Student; Attendance; Performance; Course Reports

5. User Story Repository

User Stories convert Epics and features into user-centered units of work. Required format: **As a [user], I want [capability], so that [benefit].**

ID	Epic	User Story	Priority	Points
US-01	Authentication	As a registered user, I want to log in so that I can securely access my dashboard.	High	3
US-02	Student Mgmt	As an administrator, I want to add a student so that the institution can maintain student records.	High	3
US-03	Student Mgmt	As an administrator, I want to search students so that I can quickly find a required record.	High	2
US-04	Student Mgmt	As an administrator, I want to update student information so that records remain current.	High	3
US-05	Course Mgmt	As an administrator, I want to add courses so that available academic courses are maintained.	Medium	3
US-06	Course Mgmt	As an administrator, I want to assign courses to students so that registrations are recorded.	Medium	5
US-07	Attendance	As a faculty member, I want to record attendance so that attendance records remain accurate and updated.	High	5
US-08	Attendance	As a faculty member, I want to view attendance so that I can monitor student participation.	High	3
US-09	Performance	As a faculty member, I want to enter marks so that student academic performance is recorded.	High	5
US-10	Performance	As a faculty member, I want to calculate grades so that results can be interpreted consistently.	Medium	3
US-11	Reporting	As an administrator, I want to generate academic reports so that institutional information can be reviewed.	Medium	5
US-12	Notifications	As a student, I want attendance/result notifications so that I receive important academic updates.	Medium	3
US-13	Security	As a user, I want password reset so that I can regain account access securely.	Medium	3

ID	Epic	User Story	Priority	Points
US-14	Reporting	As an administrator, I want performance reports so that academic trends can be reviewed.	Low	5

6. Acceptance Criteria

Acceptance Criteria define measurable conditions under which a User Story is considered complete and provide a clear Definition of Done.

User Story	Acceptance Criteria
Student Login	1. Login page loads successfully. 2. Registered users can log in. 3. Invalid credentials show an appropriate error. 4. Successful login redirects to dashboard. 5. Session management functions correctly.
Record Attendance	1. Faculty can select a course. 2. Faculty can select a lecture date. 3. Faculty can mark attendance. 4. Attendance is saved successfully. 5. Attendance reports can be generated.
Add Student	1. Student form loads. 2. Required details are validated. 3. Valid student data is saved. 4. Invalid/duplicate data is handled appropriately. 5. Newly added student can be searched.

7. Prioritized Product Backlog

Backlog priority is based on business value, user importance, technical dependencies and implementation complexity.

Priority	Epic	Representative User Story	Reason
High	Authentication	User Login	Required secure entry point.
High	Student Management	Add Student	Core academic record capability.
High	Student Management	Search Student	Supports quick record access.
High	Attendance	Record Attendance	Core academic workflow.
High	Performance	Enter Marks	Core academic-performance workflow.
Medium	Course Management	Add Course	Supports course workflows after foundation.
Medium	Reporting	Attendance Report	Provides management visibility.
Medium	Notifications	Attendance Alert	Improves timely communication.
Low	Dashboard / Analytics	Academic Analytics	Enhancement after core workflows.

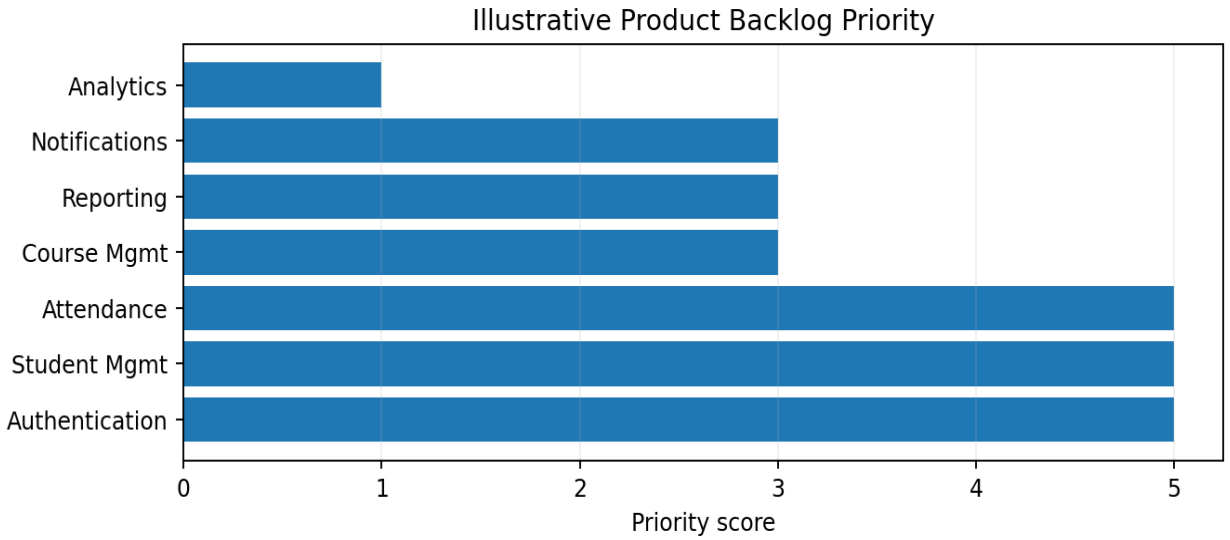


Figure 1. Illustrative backlog priority visualization; scores are planning values, not measured data.

8. Story Point Estimation

Story Points represent relative effort and complexity rather than hours. The assignment recommends Planning Poker for collaborative estimation.

Points	Complexity
1	Very Easy
2	Easy
3	Moderate
5	Difficult
8	Very Difficult
13	Highly Complex

Planning Poker

• Each member independently considers effort, complexity and dependencies.

• Estimates are revealed together and differences are discussed.

• The team agrees on a shared relative estimate.

9. Sprint Planning

The Product Backlog is divided into Sprints so the team delivers usable increments instead of attempting the entire system at once.

Sprint	Goal	Sprint Backlog / Activities	Expected deliverable
Sprint 1	Authentication & project setup	Repository initialization; registration; login; basic UI; testing	Working authentication foundation
Sprint 2	Student Management	Add; Search; Update; Delete Student	Working student-management module
Sprint 3	Course & Attendance	Add Course; Assign Courses; Record Attendance; Attendance Reports	Working course + attendance workflows
Sprint 4	Performance, Reporting & Notifications	Enter/Update Marks; Grades; Result Sheets; Reports; Notifications; testing	Integrated academic-management increment

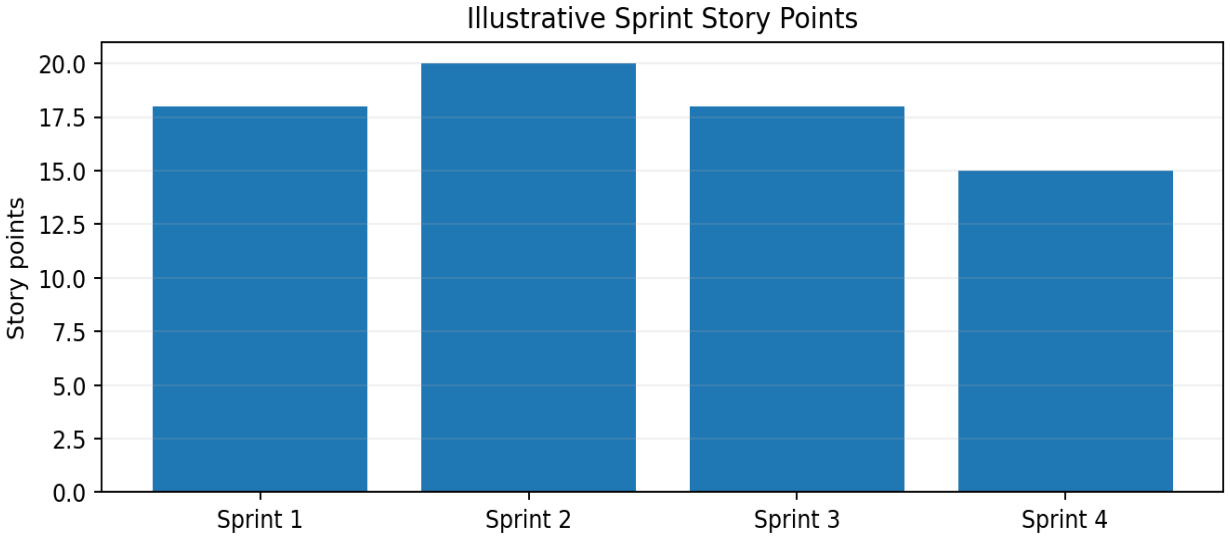


Figure 2. Illustrative sprint story-point distribution; the source assignment does not prescribe numerical sprint totals.

10. Scrum Board

A Scrum Board makes work visible and supports movement of User Stories through the development workflow.

Backlog	To Do	In Progress	Testing	Completed
Password Reset	Add Student	Search Student	Attendance Report	Repository Setup
Course	Attendance	Notifications	Login	Basic UI
Analytics	Enter Marks	Result Sheet	Student Update	Registration

Board flow: Backlog → To Do → In Progress → Testing → Completed. A story should reach Completed only after its acceptance criteria are satisfied.

11. Scrum Ceremonies

Ceremony	Purpose	Project activity
Sprint Planning	Plan the Sprint	Select prioritized User Stories; estimate effort; define Sprint Goal.
Daily Scrum	Synchronize the team	What did I complete yesterday? What will I do today? What challenges am I facing?
Sprint Review	Inspect the increment	Demonstrate completed features and collect stakeholder/user feedback.
Sprint Retrospective	Improve the process	Discuss what went well, challenges encountered and improvements for the next Sprint.

12. Agile Framework Comparison

The assignment asks for a comparison of Scrum, Kanban, Extreme Programming (XP), Lean Development, LeSS, SAFe and Scrum@Scale, followed by a recommendation based on project size and complexity.

Framework	Main emphasis	Fit for this project
Scrum	Time-boxed Sprints, roles and ceremonies	Strong fit — matches the assignment's Sprint, backlog and ceremony structure.
Kanban	Continuous flow and work visualization	Useful for maintenance/support; less structured for this Sprint-based mini-project.
XP	Engineering practices and frequent feedback	Useful complementary practices, especially testing and integration.
Lean Development	Waste reduction and value delivery	Useful mindset for prioritization and avoiding unnecessary work.
LeSS	Scaling Scrum across multiple teams	More suitable when multiple Scrum teams coordinate on one product.
SAFe	Enterprise-scale Agile coordination	Designed for larger organizations and portfolios; more than needed here.

Framework	Main emphasis	Fit for this project
Scrum@Scale	Scaling Scrum across teams	Useful for larger multi-team environments; unnecessary for the initial project.

Recommendation: Scrum is the most suitable primary framework for this Unit-2 project because the assignment explicitly organizes work around Product Backlog, User Stories, Sprints, Scrum Board and Scrum ceremonies. XP practices can complement Scrum, especially for testing and integration.

13. Project Timeline & Release Roadmap

Stage	Focus	Output
Unit 2 — Agile Planning	Vision → Roles → Epics → Stories → Acceptance Criteria	Complete Agile planning baseline
Sprint 1	Authentication & setup	Secure access foundation
Sprint 2	Student Management	Student CRUD/search workflows
Sprint 3	Course + Attendance	Course assignment and attendance workflows
Sprint 4	Performance + Reporting + Notifications	Integrated academic increment
Post-Unit Expansion	Further refinement, testing and engineering practices	More complete production-ready platform

14. Burndown Chart — Tracking Method

The Unit-2 deliverables include a Burndown Chart. Since the source assignment does not provide actual daily completion data, this section defines the tracking structure rather than fabricating results.

Day	Ideal remaining points	Actual remaining points
Day 1	Sprint total	Record after planning
Day 2	Target remaining	Record actual
Day 3	Target remaining	Record actual
…	…	…
Final day	0	Record actual remaining work

15. Definition of Done

• Acceptance criteria are satisfied.

• Feature is implemented according to the User Story.

• Relevant testing has been completed.

• Known blocking defects are resolved.

• Documentation is updated where required.

• Feature is demonstrated during Sprint Review.

16. Unit-2 Submission Checklist

■ **01** Agile Project Planning Report

■ **02** Product Vision Document

■ **03** Scrum Team Structure

■ **04** Scrum Roles and Responsibilities Document

■ **05** Epic Documentation

■ **06** User Story Repository

■ **07** Acceptance Criteria Document

■ **08** Prioritized Product Backlog

- **09** Story Point Estimation Sheet
- **10** Sprint Planning Document
- **11** Sprint Backlog
- **12** Scrum Board
- **13** Burndown Chart
- **14** Sprint Review Report
- **15** Sprint Retrospective Report
- **16** Agile Framework Comparison Report
- **17** Project Timeline and Release Roadmap
- **18** Complete Agile Documentation for the project

17. Conclusion

This Unit-2 Agile mini project converts the Unit-1 Student Record and Academic Management Platform into an actionable Agile development plan. The project is decomposed into seven Epics, User Stories are prioritized using business value and user importance, Story Points support relative estimation, and the work is organized into four example Sprints. Scrum roles, ceremonies, a visible Scrum Board, acceptance criteria, framework comparison and release planning provide the foundation for incremental software development.

Source note: This report is grounded in the supplied “UNIT-2-mini-project.pdf”. Numerical chart values labeled illustrative are planning examples added for visual understanding; they are not presented as measured project data.